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A Competing Risks Analysis Assessing Predictors of Neurologic and Non-Neurologic Death in Patients With Brain Metastasis Initially Treated With Upfront Stereotactic Radiosurgery Without Whole-Brain Radiation Therapy

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Purpose/Objective(s): Recent clinical trials suggest rates of neurologic death are higher in patients with brain metastasis initially treated with stereotactic radiosurgery (SRS) alone as compared to those treated with SRS, plus subsequent whole-brain radiation therapy (WBRT). In this study we attempt to discern the factors predictive of neurologic death in patients with brain metastasis treated with upfront SRS alone.

Materials/Methods: Six hundred thirty-seven of 738 patients at an academic center were treated initially with SRS and were either alive or had complete information regarding cause of death at the time of analysis. Neurologic death was defined as death from progressive neurologic dysfunction irrespective of systemic disease status. Univariate analysis identified predictive covariates with a P value $\leq .1$ for consideration of inclusion into a multivariate (MVA) Cox proportional hazards model. Competing risks analysis was performed to estimate the subdistribution hazard ratios (HRs) for neurologic death and non-neurologic death for each predictor.

Results: Of 738 patients treated with upfront SRS alone, 636 (86.2%) patients were deceased at time of analysis. Of these patients, the cause of death was neurologic in 226 (35.5%), non-neurologic in 309 (48.6%), and unknown in 101 (15.9%). Predictors of neurologic death identified on univariate analysis included histology ($P = .03$), minimal SRS dose ($P = .03$), number of brain metastases ($P < .0001$), systemic disease status ($P = .003$), KPS ($P < .0001$), and post-SRS RTOG symptom grade ($P < .0001$). Multivariate competing risks analysis identified an increased hazard of neurologic death in patients with a higher post-SRS RTOG symptom grade ($P < .0001$) of 2 (HR = 1.9, 95% CI 1.3-2.7), 3 (HR = 3.1, 95% CI = 1.8-5.4) or 4 (HR = 5.8, 95% CI 3.2-10.3), and melanoma histology (HR = 2.0, 95% CI 1.1-3.6, $P = .03$). Progressive systemic disease was associated with an increased hazard of non-neurologic death (HR = 1.8, 95% CI 1.4-2.4, $P = .0001$).

Conclusion: In patients with brain metastasis treated upfront with SRS alone, higher post-SRS RTOG symptom grade was the dominant factor in predicting neurologic death. This finding validates the historic notion that metastatic patients with symptomatic intracranial disease have an inferior prognosis secondary to an increased hazard of neurologic death. Conversely, the presence of progressive systemic disease was the primary predictor of non-neurologic death. Careful consideration of the effect of classic prognostic factors on terminal events may be useful for defining high-risk disease.

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Reirradiation of the Spinal Cord: Efficacy and Toxicity

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Purpose/Objective(s): Tumor recurrence in the spinal and paraspinal area often warrants reirradiation; however, limited data is available on the efficacy and potential consequences of reirradiating the spine. This study aimed to determine the potential benefits and adverse effects associated with reirradiating the spine when at least one course of RT is stereotactic radiosurgery (SRS).

Materials/Methods: One hundred eighty-two patients and 372 re-irradiated spine lesions were retrospectively reviewed in this IRB approved analysis. Each patient received at least one course of SRS at our institution from June 2001 to December 2013. Electronic medical records of clinical exams, and CT/ MRI were evaluated. Primary endpoints were pain response, neurological response, radiographic tumor control and the development of adverse effects (including radionecrosis, development of vertebral compression fractures (VCFs), neurological deficits and death).

Results: Survival after last RT treatment was a median of 6.9 months (range, 0 days to 6.7 years). At the time of analysis, 131 patients (72%) were deceased. Median follow-up time was 3.4 months (range, 1 day to 96 months), mean follow-up time was 13 months. Time between courses of RT was an average of 21 months (median, 9.8 months). The average cumulative dose delivered to a lesion was 46 Gy (30 Gy/10 fx standard RT & 16 Gy/1 fx SRS). For SRS planning the dose constraint used was 10 Gy to 10% with a maximum point dose of 14 Gy. Cumulative dose delivered to a lesion ranged from 14 Gy-107 Gy. Stereotactic radiosurgery dose range was 4-24 Gy (in 1-4 fx) and standard RT dose range was 8-66 Gy (in 1-25 fx). Subdividing this for the region in the spinal cord and that below the spinal cord showed that the C1-L2 region was treated to a median of 50 Gy (range, 14 – 100 Gy) and for L3-Sacrum median = 47 Gy (range, 27 - 107 Gy). [Details of these high dose patients (100 Gy and 107 Gy) will be provided further]. Overall pain response rate was 81% (63% PR, 18% CR), stable in 5%, and progressed in 14%. Neurological improvement occurred in 82% (55% PR, 27% CR), remained stable in 9% and progressed in 9%. Radiographic tumor control was observed in 71% (47% stable, 23% partial, 1% CR). Twenty-nine percent experienced tumor progression. Adverse effects occurred in 11 patients. Radionecrosis and myelomalacia each occurred in one patient (0.55%). Three patients (1.65%) experienced dysphagia/odynophagia. Sensory changes, weakness or radiculopathy occurred in five patients (2.75%). One death was due potentially to radiation related complications (0.55%). Seventy-seven vertebral compression fractures (VCFs) were observed, 22 (6%) of which may be attributed to RT.

Conclusion: Our results demonstrate that reirradiation achieves favorable response rates. If recommended dose constraints to the spinal cord with SRS are followed (10 Gy to 10% of the spinal cord), complication rates are low.

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Patient, Tumor, and Dosimetric Predictors of Temporal Progression in Visual Acuity Following Proton Beam Radiation Therapy for Uveal Melanoma

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Purpose/Objective(s): To discern patient, tumor, and critical structure dose-volume parameters affecting visual acuity outcomes as a detailed temporal analysis after proton beam radiation therapy (PBRT) for uveal melanoma (UM).

Materials/Methods: Using a prospectively maintained database, UM patients treated with PBRT (1994-2010, n = 602) were stratified according to pretreatment best corrected VA (BCVA) in the treated eye of $\geq 20/40$